



Explainable ensemble-based machine learning model for polycystic ovary syndrome detection using hybrid feature selection

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Abstract

Polycystic Ovary Syndrome (PCOS) is a common endocrine condition that needs accurate diagnosis for effective management. It involves the presence of numerous immature follicles in the ovaries, which can interfere with healthy ovulation and lead to hormonal imbalances and other health issues. Consequently, it is essential to establish a PCOS detection system that is both precise and timely to lower complications. In the current literature, Machine Learning (ML) models have demonstrated their efficacy in detecting PCOS. However, the accurate and early detection of PCOS requires the precise identification of key features. This paper proposes a hybrid framework for PCOS prediction that combines ensemble learning and feature selection. The proposed methodology integrates Genetic Algorithm (GA), Mutual Information (MI), and Boruta feature selection techniques to identify the most informative clinical and hormonal features. In addition, to facilitate a comparative evaluation of prediction performance, a variety of base and ensemble classifiers were trained with selected features. The hybrid feature set improved diagnostic accuracy and generalizability across models, establishing a comprehensible and effective method for PCOS identification that is suitable for clinical decision support. Additionally, SHAP-based feature interpretation is performed to assess the contributions of each feature. The proposed method is evaluated on a publicly available PCOS dataset. It exhibits superior performance compared to several existing approaches, achieving an accuracy of over 94% on all different combinations of feature sets and XGBoost.

Keywords Boruta · Genetic algorithm · Polycystic ovary syndrome (PCOS) · Machine learning · Ensemble learning

1 Introduction

Polycystic Ovary Syndrome (PCOS) is a complex and multifactorial endocrine disorder in approximately 5–20% of women of reproductive age all over the world. PCOS has a spectrum of manifestations that include hyperandrogenism,

ovulatory dysfunction, and polycystic ovaries. Escobar-Morreale [1] characterized PCOS as a dynamic interaction of complex environmental, genetic, and neuroendocrine factors, and thus the challenge in diagnosis and management. Besides reproductive issues, PCOS is linked to chronic metabolic abnormalities [2, 3], including insulin resistance, obesity, and dyslipidemia, which increases the need for early diagnosis.

To address the diagnostic challenges associated with PCOS, several researchers proposed intelligent systems to analyse clinical data, incorporating ML algorithms. The work in [4] introduced a Smart Polycystic Ovary Syndrome Diagnostic System (SPOSDDS) utilizing ML for the diagnosis of PCOS based on clinical and hormonal features. Their approach demonstrated the capability of Artificial Intelligence (AI) in accurately and timely identifying PCOS in patients. These kinds of intelligent diagnosis tools are beneficial in regions where there is a limited number of expert medical professionals.

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Besides affecting reproductive health, PCOS is strongly linked to the onset of type 2 diabetes, cardiovascular diseases, and cerebrovascular conditions. Authors in [5] underscore the notable prevalence of metabolic comorbidity in women with PCOS, including those of normal weight. The study highlights the importance of early detection in preventing the progression of chronic diseases. Danaei Mehr and Polat [6] performed a comprehensive comparative analysis of various ML algorithms and feature selection methods for diagnosing PCOS utilizing clinical datasets. Ge et al. [7] also proposed applying intelligent feature selection and model simplification for clinical prediction tasks.

The missing values in real-world clinical datasets often lead to incorrect predictions and poor model performance. Through an extensive examination of imputation methodologies tailored for healthcare data Benhamza et al. [8] emphasized the importance of imputation methods. Their findings demonstrated the importance of proper data preparation as a prerequisite for efficient ML pipelines.

Recent developments in AI and medical imaging underscore the importance of model interpretability in clinical diagnosis. Akbar et al. [9] emphasized the importance of openness in medical decision-making, who used 3D dynamic contrast-enhanced Magnetic Resonance Imaging (MRI) to illustrate the use of explainable AI for breast cancer prediction. They demonstrated that adding explainability to AI models builds confidence and also helps physicians to explain the reasoning behind diagnostic forecasts.

Recently, healthcare research has concentrated on advanced ML algorithms due to its capability to identify intricate, nonlinear patterns in patient data that conventional statistical methods frequently overlook [10]. ML methodologies assist in identifying the most significant features within high dimensional datasets, particularly when coupled with robust feature selection algorithms. The proposed work introduces a comprehensive framework that integrates several feature-selection methods and ensemble learning to improve the precision of PCOS predictions while preserving clinical interpretability and reliability. The primary contributions of the proposed work are as follows.

- Compared several feature selection methods, such as Mutual Information (MI), Genetic Algorithm (GA), and Boruta, followed by a union strategy and reduction of redundancy through correlation filtering.
- A comprehensive analysis of several conventional ML classifiers for PCOS detection.
- Proposed an ensemble ML classifier using hybrid feature sets and identifying the appropriate base and ensemble models based on diverse evaluation metrics.
- Employed model interpretability techniques to assess the significance of features in prediction.

The following are the remaining sections of the article. A summary of the existing works on PCOS detection is provided in Sect. 2. A detailed explanation of the dataset and proposed methodology is given in Sect. 3. The experiments that were conducted and the analysis of the results obtained are detailed in Sect. 4. Finally, Sect. 5 concludes the work by outlining future directions.

2 Related works

The current literature presents various methodologies for detecting PCOS [11]. Several methodologies employing ML models have been proposed in the literature to enhance PCOS identification. This section presents an analysis of the principal methods for PCOS detection.

Tiwari et al. [4] proposed a smart decision-making model, Random Forest (RF) and Decision Tree (DT) for PCOS detection. This method identified significant features using feature selection on a combination of clinical and symptom-based features. The main benefit of this method is that it combines symptomatology and predictive modeling in a way that gives better categorization results. However, its actual implementation is limited by dataset biases.

Zhu et al. [5] investigated the long-term metabolic comorbidities associated with PCOS, such as type 2 diabetes, coronary heart disease, and stroke. In which the longitudinal epidemiological data and statistical models evaluated the associations between PCOS and systemic health risks. The key benefit of this is that it emphasizes the importance of public health and highlights the significance of identifying and treating PCOS at an early stage. PODBoost, an interpretable AI model introduced by Moral et al. [12], using a grey wolf optimization-based feature selection for PCOS classification. Grey wolf optimization, a metaheuristic algorithm, facilitates an adaptive and dynamic search for the best sets of features, thereby improving model accuracy and convergence. However, the application of metaheuristics is generally computationally costly and sensitive to parameter configuration, makes it unsuitable for real-time applications.

Authors in [13] introduced a PCOS detection model using Support Vector Machine (SVM), Decision Tree (DT), and K-Nearest Neighbor (KNN) classifiers, along with classic feature ranking methods. As the process is straightforward, it is easily applicable in clinical settings. However, traditional methods for ranking features may not adequately account for multivariate interactions. Authors in [14] used multiple ML models, like DT, NB, and SVM, with basic correlation-based feature selection to diagnose PCOS. The most significant advantage is that the feature selection method is straightforward to utilize, even for non-technical

clinical scientists. However, simple correlation-based filtering cannot identify nonlinear relations and may exclude features that are only helpful when combined with other features.

Hassan and Mirza [15] performed a comparative analysis of the ML models, such as RF, SVM, and KNN, for PCOS diagnosis with both filter and wrapper feature selection methods. Their research illustrates how classifier performance varies depending on the strategy employed for feature selection. The strength is that it is a complete handbook for model–feature selection pair selection. Yet, the drawback is that there is no automation in selecting the optimal combination, and the tuning requires manual operation as well as extensive experimentation. Emara et al. [16] proposed a stacked learning framework to classify PCOS using a combination of Adaptive Synthetic Sampling (ADASYN) for class imbalance handling, Boruta for feature selection, and a stacked ensemble classifier. The model effectively addressed the skewed distribution of features in the dataset, focusing on highly relevant features to achieve superior classification performance. One of the notable advantages of their approach is robustness across multiple datasets. However, it is computationally expensive for real-time applications due to the multi-step preprocessing pipeline.

Alam Suha and Islam [17] designed an enhanced stacking ensemble model for detecting PCOS, utilizing both clinical and symptomatic data. They conducted feature reduction using Principal Component Analysis (PCA) and Recursive Feature Elimination (RFE). The model performed exceptionally well in identifying the most critical predictive features and mitigating overfitting by employing ensemble learning. However, the dependence on PCA may diminish the interpretability of features. Panjwani et al. [18] proposed an optimized ML framework for early PCOS detection using univariate feature selection methods like Chi-square and SelectKBest, paired with hybrid classifiers. Despite its high performance, the absence of complex multivariate feature selection may limit insights into feature interactions.

Uddin et al. [19] conducted a comparative analysis between traditional classifiers, including SVM and Logistic Regression (LR), and ensemble models such as XGBoost and RF, incorporating advanced feature selection techniques. The study employed statistical methods, such as the Chi-square test and MI, to reduce dimensionality. Their results showed excellent performance in ensemble models. El-Sayed et al. [20] improved PCOS classification by employing a metaheuristic optimization algorithm in conjunction with Boruta for feature selection. The proposed approach was integrated into a hybrid classifier, yielding improved accuracy and reduced dimensionality. It proved to be more effective in investigating the feature space with greater adaptability than conventional optimizers. Although

new, the optimization approach may need to be validated on different datasets or medical circumstances.

According to Taher et al. [21], a multi-layer stacking ensemble can be used to find PCOS by employing simple feature selection methods like MI and Pearson correlation. There were three base learners in their architecture: NB, RF, and LR. The meta-classifier was a Multi-Layer Perceptron (MLP). The stacking design is strong because it is simple and can be divided into smaller parts. However, it may not be able to capture complicated feature interactions since the feature selection is too simplistic. The work by Gupta et al. [22] compared AdaBoost, XGBoost, and CatBoost as boosting algorithms for classifying PCOS. The study did not employ explicit feature selection; instead, it demonstrated how boosting algorithms can be applied to high-dimensional data. Although boosting methods have shown strong performance, the absence of a feature selection strategy or interpretability mechanism presents a constraint.

Nair et al. [23] focused on the stacked generalization approach to binary classification for health diagnostics, including PCOS. The model used different base learners, such as SVM and DT, along with a meta-learner. This made the classification more accurate than using just one model. While stacking improved generalizability, it lacks attention to feature reduction or interpretability.

Several studies have examined the feasibility of utilizing ultrasound imaging and ML for PCOS detection. Adiwijaya et al. [24] proposed a method to detect follicles in ultrasound scans, which enables doctors to diagnose issues by examining the images. Wisesty et al. [25] enhanced classification accuracy in image-based PCOS detection using an innovative backpropagation method. Complementary studies on image processing techniques include digital image compression using wavelet-SVD in combination with graph coloring [26] and follicular segmentation via particle swarm optimization (PSO) [27].

Silambarasan et al. [28] proposed two different methodologies for PCOS classification, utilizing both text and image-based datasets. Relevant features from the text-based dataset were extracted utilizing Recursive Feature Elimination (RFE), L1 regularization, tree-based methods, and Principal Component Analysis (PCA). Following, SVM classifier was used to classify the features. In contrast, DenseNet architecture was proposed for feature extraction and classification in the image-based dataset.

Detecting PCOS is difficult due to the prevalence of noise in ultrasound images. Therefore, to improve the quality of images, [29] proposed a denoising filter that eliminates anomalies from PCOS ultrasound images in the time–frequency domain. Rajput et al. [30] evaluated the efficacy of different ML algorithms, such as XGBoost, DT, Artificial Neural Network (ANN), and RF, in the classification

of PCOS, according to clinical and laboratory parameters. The Sine Cosine Algorithm (SCA) was employed in their work to optimize feature selection, thereby improving the accuracy and efficacy of the model. A summary of relevant existing literature utilizing ML for PCOS detection, which served as the motivation for the proposed work, is presented in Table 1.

Recent advancements in explainable artificial intelligence (XAI) have underscored interpretability as a critical issue in medical diagnosis. Hagrais [34] stressed that AI systems that people can understand are important for trust and openness, especially in sensitive areas like healthcare. Islam et al. [35] did a systematic review of XAI, looking at different methods in different fields and stressing how important it is to be able to understand, to choose, and check models. Comparison studies like those of Vij and Nanjundan [36] further discussed the performance of various explanation strategies. Specifically targeting PCOS, Witchel et al. [37] wrote about the multifaceted presentation of the disorder in adolescents, emphasizing the issues of diagnosis that require explainable models.

3 Methodology

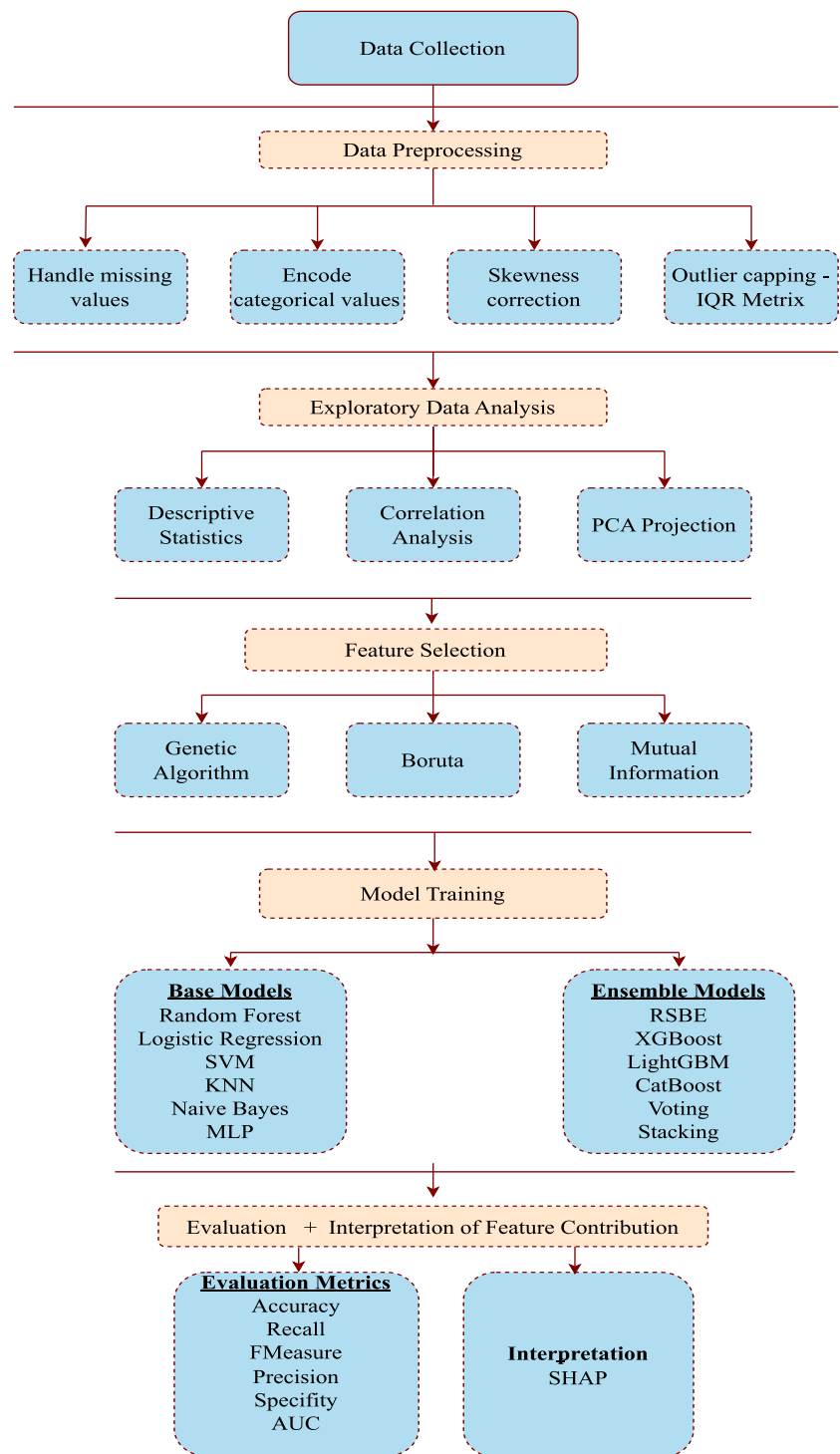
The proposed methodology for PCOS identification comprises a series of phases, starting with preprocessing and concluding with evaluation. Figure 1 shows the overall workflow in the proposed work for PCOS detection. A detailed description of the dataset utilized and the various steps included in the proposed work is provided below.

3.1 Dataset description

The performance of proposed method is evaluated on a publicly accessible PCOS dataset [38] from Kaggle. The dataset provided clinical information on 541 women, of whom 177 had a PCOS diagnosis and 364 did not. The dataset offers 43 features, including Clinical measurements (e.g., blood pressure, BMI), hormonal indicators (e.g., Follicle Stimulating Hormone (FSH), Luteinizing Hormone (LH), Anti-Müllerian Hormone (AMH)), symptomatic information (e.g., acne, hair growth), and ultrasound-derived variables (e.g., follicle count and size). The binary target variable indicates if PCOS, as determined by medical professionals, is present or not.

Table 1 Summary of existing works on PCOS detection

Paper	Feature selection	Sampling	Classifier	Remarks
Tiwari et al. [4]	Pearson correlation	SMOTE	SVM, DT, RF, LR, AdaBoost, KNN, GB, XgBoost, CatBoost, LDA, QDA	The varying classifier results demonstrate that effective preprocessing and feature selection enhance the results
R. et al. [31]	Correlation ranking, Chi square test and recursive feature elimination, PSO and flashing firefly (FF)	SMOTENC	RF and XGBoost	Research on effective feature selection and ML classifiers can improve the diagnostic model. New PCOS datasets from different areas and demographics would also assist in building a more accurate prediction model
Bharati et al. [14]	Filtering based univariate	–	GB, RF, LR and hybrid RF and LR	A limited comparative investigation has been conducted. Furthermore, multiple hybrid approaches can be devised to enhance the classification accuracy of ML algorithms
Zig-arelli et al. [32]	PCA	–	CatBoost	A larger dataset from different ethnicities may improve the performance
Khan Inan et al. [33]	ANOVA and chi-square	SMOTE and edited nearest neighbour	Extreme GB, XGBoost	More comprehensive hyperparameter tuning of ML algorithms and enhanced feature selection can achieve enhanced performance
Proposed work	GA, Boruta, MI, union features followed by correlation filtering	SMOTE	RF, LR, SVM, KNN, NB, MLP, RSBE, XGBoost, LightGBM, CatBoost, voting, stacking	A more extensive feature analysis has been conducted. Additionally, it has explainability for interpreting feature contributions using SHAP

Fig. 1 Proposed workflow for PCOS classification

3.2 Data pre-processing

To pre-process the PCOS dataset for predictive modeling, an extensive data pre-processing pipeline was utilized. First, two independent datasets—one with clinical records and the other with hormonal profiles—were concatenated using a common identifier to have each patient represented with

a complete set of features. For handling missing values, median imputation was performed for all numerical columns. This approach was chosen instead of mean imputation because of its stability under skewed distributions and outliers, which were common in variables such as Beta-Human Chorionic Gonadotropin (Beta-HCG) values and cycle duration.

Class imbalance in the dataset is addressed using SMOTE. This ensured the balancing of positive and negative examples of PCOS, so that the models were not biased towards the majority class, as shown in Fig. 2a and b.

Treatment of outliers was done through percentile capping at the 1st and 99th percentiles, which served to reduce the impact of outlier values without eliminating possible useful information. All the features were numericized, and any entries that could not be converted were imputed in the cleaning process. Categorical variables, such as blood group and marital status, were encoded using one-hot encoding, depending on the model's demands. Numerical features were standardized using StandardScaler, converting them to a mean of zero and a standard deviation of 1. This was especially necessary for algorithms such as KNN, SVM, and MLP, which are sensitive to feature scales.

To highlight the influence of capping, a few of the most significant features with high skewness were plotted both before and after capping. Figure 3a illustrates that the distribution of the feature 'I_betaHCGmIU/mL' was highly right-skewed, with outliers. However, after the capping process, the tail portion of the distribution decreased significantly, resulting in a more centered and compact distribution. Figure 3b shows the 'Cycle_lengthdays' feature, which had a wider spread at first. But after capping, it became more compact and symmetrical. Figure 3c shows how capping affects 'Marriage_Status_Yrs'. Lastly, Fig. 3d for 'Vit_D3_ngmL' shows a big change from a sparse distribution with very high correct values to a narrower range after capping. These conversions not only reduced skewness and the effect of outliers but also helped with better feature scaling and

model convergence during training. To protect data, the unnecessary identifiers 'SI_No' and 'Patient_File_No' were removed.

3.3 Exploratory data analysis

To gain a detailed understanding of the dataset, the correlation between features and the target variable, exploratory data analysis was conducted in the proposed work. Then, using the correlation heatmap shown in Fig. 4, the relationships between features were determined. According to this, there were strong links between 'BMI' and other measures of obesity, such as 'Weight_Kg'. Also, it shows the relation between certain hormones, such as 'FSHmIU/mL' and 'FSHLH'. These insights were essential for guiding feature selection and understanding multicollinearity.

PCA was used to show data separability in two dimensions, as shown in Fig. 5. The PCA plot also indicated a moderate degree of clustering between the PCOS and non-PCOS groups. This supports the idea that the features we chose had important information that allowed us to differentiate them.

3.4 Feature selection

The proposed method used a range of feature selection techniques, including hybrid, filter-based, and wrapper-based approaches. These consist of Boruta, MI, GA, and a hybrid union strategy that combines the advantages of all three. A strong feature engineering pipeline is made possible by the

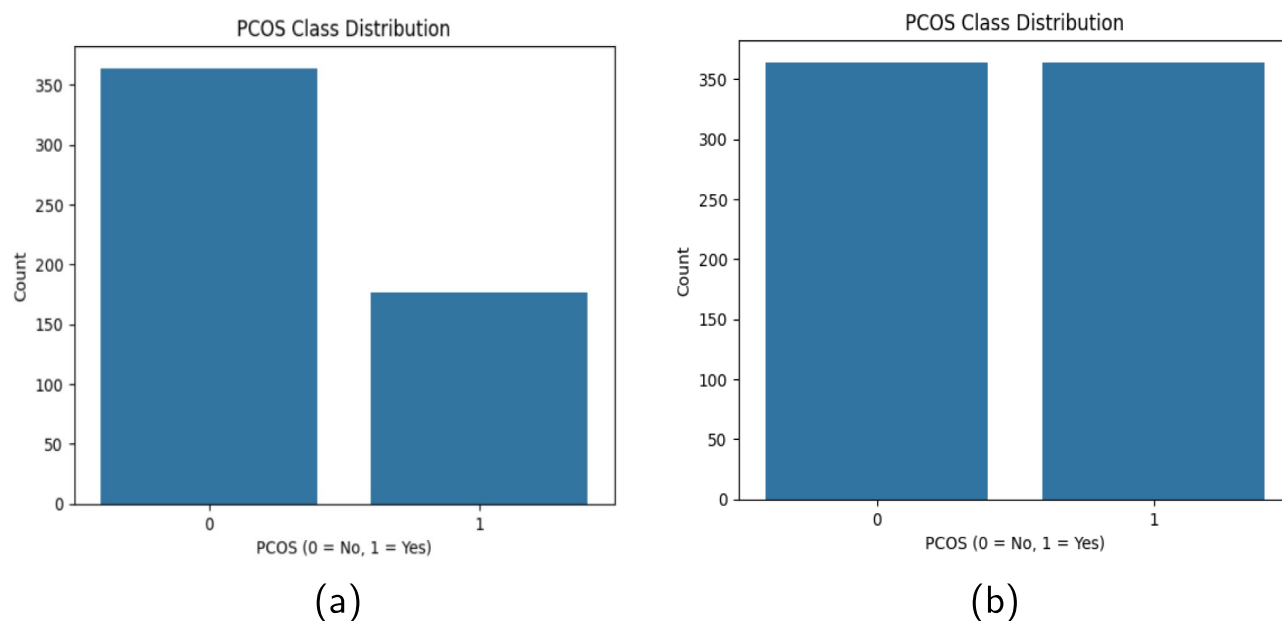


Fig. 2 PCOS distribution: **a** Before applying SMOTE, **b** After applying SMOTE

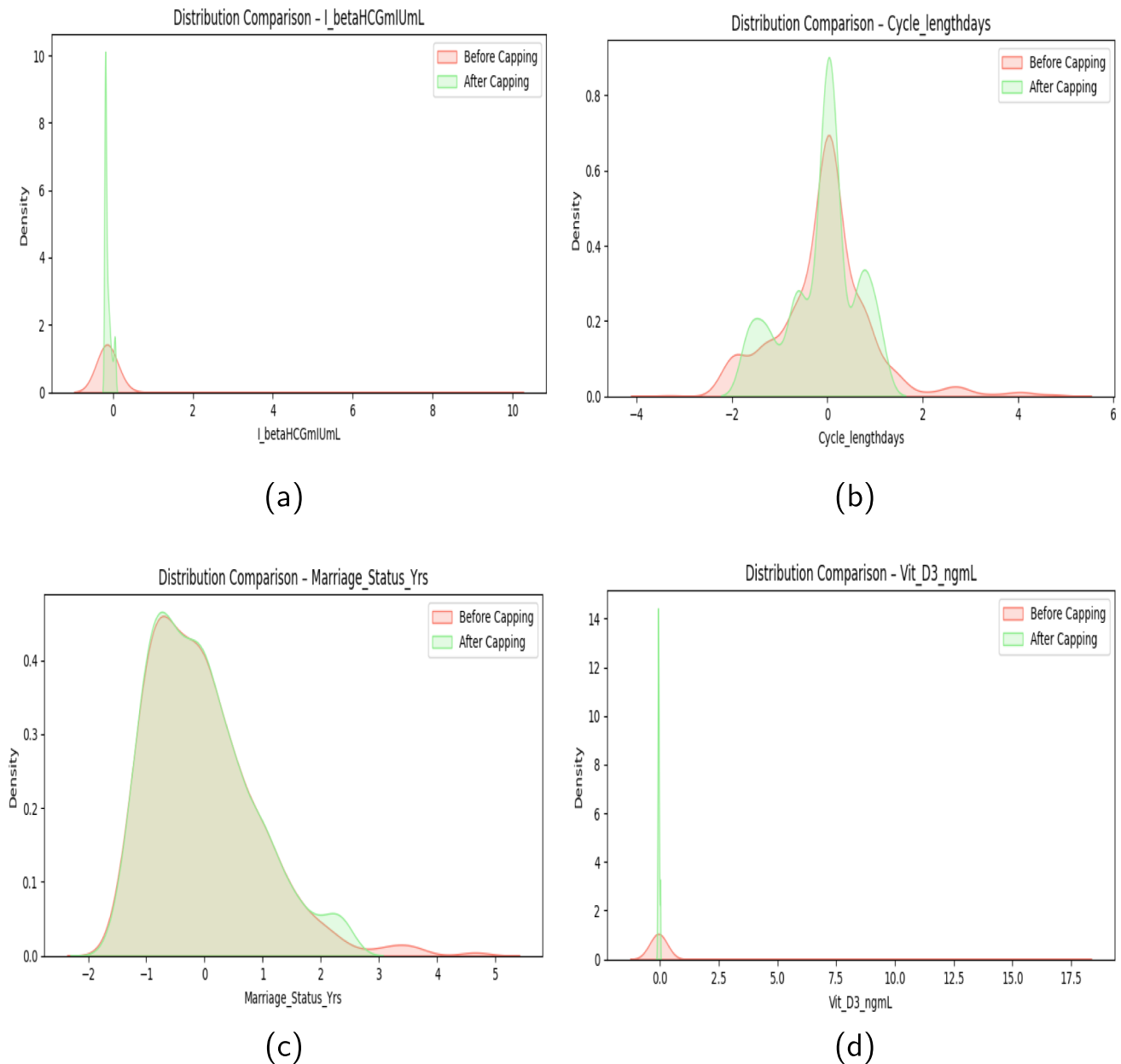


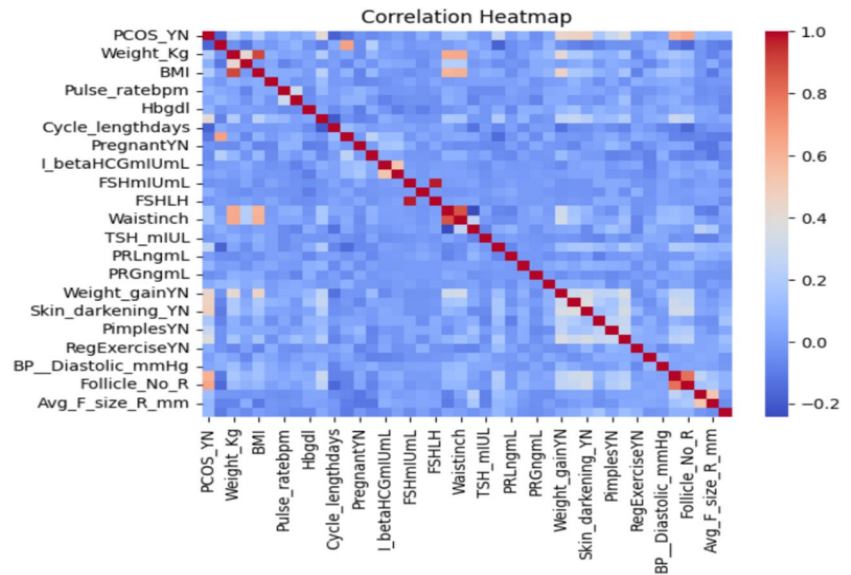
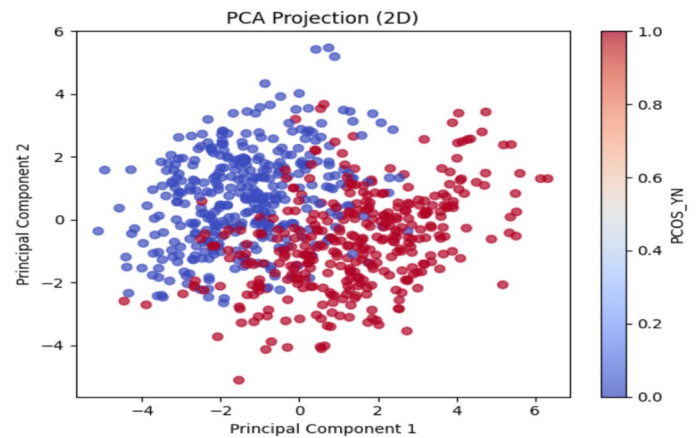
Fig. 3 Effects of capping on various features: **a** I_betaHCGmIUmL, **b** Cycle_lengthdays, **c** Marriage_Status_Yrs, **d** Vit_D3_ngmL

distinct perspectives that each technique offers on identifying informative features.

The GA was applied with the DEAP library. GA simulates natural evolutionary operations to find optimal sets of features. A set of random binary chromosomes for various feature combinations was evolved through selection, crossover, and mutation operations. The fitness of each chromosome was calculated as the average of F1-score over a RF classifier with 5-fold cross-validation. The best-performing chromosome's corresponding set of features was chosen for subsequent training of the model.

In the MI method, statistical dependence between every feature and the target variable was measured. Features with the highest MI values were selected, revealing both linear and non-linear relationships with the target. The 18 most relevant features with highest MI values were selected to balance model simplicity and accuracy.

Boruta was used as a wrapper-based feature selector constructed on top of RF. It iteratively matched the importance of true features with that of randomly shuffled shadow features. Features that consistently outperformed their shadows across iterations were retained. This filtered only highly

Fig. 4 PCOS correlation heatmap**Fig. 5** PCA projection

relevant features and dropped weak or irrelevant ones with statistical stringency.

Additionally, in the proposed method, a union of the features selected by GA, MI, and Boruta was formed to leverage the strengths of all three techniques. This combined feature space consisted of all variables considered significant by at least one of the techniques. To minimize the redundancy that might result from such a union, multicollinearity was subsequently addressed by eliminating highly correlated features based on a correlation threshold.

3.5 Model training

In the proposed work, a comparative analysis of various base and ensemble ML models is evaluated using the different types of feature sets. In the proposed work, as part of the base ML models, the analysis included RF, LR, SVM,

KNN, NB, and MLP models. The prediction performance is improved using various ensemble learning models, including Random Subspace with Bagging Ensemble (RSBE), XGBoost, LightGBM, CatBoost, Voting, and Stacking. The RSBE model is constructed utilizing ten Gaussian NB models, each trained on bootstrapped samples comprising 70% of the features. The final prediction is determined through majority voting among all NB models. The CatBoost model is configured with 300 iterations, a learning rate of 0.03, a tree depth of 6, and 64 split points. In the case of stacking, the NB, KNN, and RF classifiers are used as level-0 learners, whereas the LR classifier is used as the level-1 learner. The voting classifier model used soft voting from LR, RF, and SVC. Algorithm 1 shows the main steps that were followed in the proposed study.

Require: Dataset d comprising of features f and target variable y

Ensure: Optimal model with best predictive performance

1: **Step 1: Data Preprocessing**

1. Handle missing values and convert the data into numeric.
2. Use the Interquartile Range (IQR) approach to apply outlier capping.
3. Apply *StandardScaler* for feature normalization.

2: **Step 2: Feature Selection**

1. Apply Genetic Algorithm to create a feature subset based on F1-score.
2. Apply Mutual Information and select top-ranked features.
3. Apply Boruta and select features utilizing Random Forest importance ranking.
4. Compute Hybrid Feature Set: $f_{hybrid} = f_{GA} \cup f_{MI} \cup f_{Boruta}$.

3: **Step 3: Model Training**

1. Split the data into 70:30 training and testing datasets.
2. Train the following base models: {KNN, Logistic Regression, MLP, Naive Bayes, Random Forest and SVM} for all the feature sets.
3. Train the following ensemble models: {LightGBM, RSBE, Stacking, Voting and XGBoost} for all the feature sets.

4: **Step 4: Evaluation**

1. Predict on the test dataset.
2. Compute metrics: Accuracy, F1-score, Sensitivity, Specificity and Precision.
3. Generate confusion matrix for every model and plot ROC curves.

5: **Step 5: Model Comparison**

1. Deduce the best performing model.
2. Compare the outcome with base study.

6: **Step 6: Interpretation of Influencing Factors**

1. Assess the feature contribution using SHAP.

7: **return** The model combination and feature selection method that performs best.

Algorithm 1 Proposed Framework for PCOS Prediction

4 Experiments, results and analysis

A comprehensive discussion of the several experimental setups and evaluation methods used are discussed in this section. Additionally, we highlight the observed outcomes across several feature sets using a range of ML models. This section also addresses the illustration of feature interpretability.

4.1 Training and evaluation

In the proposed work, various classification models are trained using different feature sets. This includes classic baseline models like RF, LR, SVM, KNN, NB, and a MLP, in addition to ensemble models like Voting, Stacking, RSBE, XGBoost, CatBoost, and LightGBM. All models were trained on stratified data splits with a ratio of 70:30,

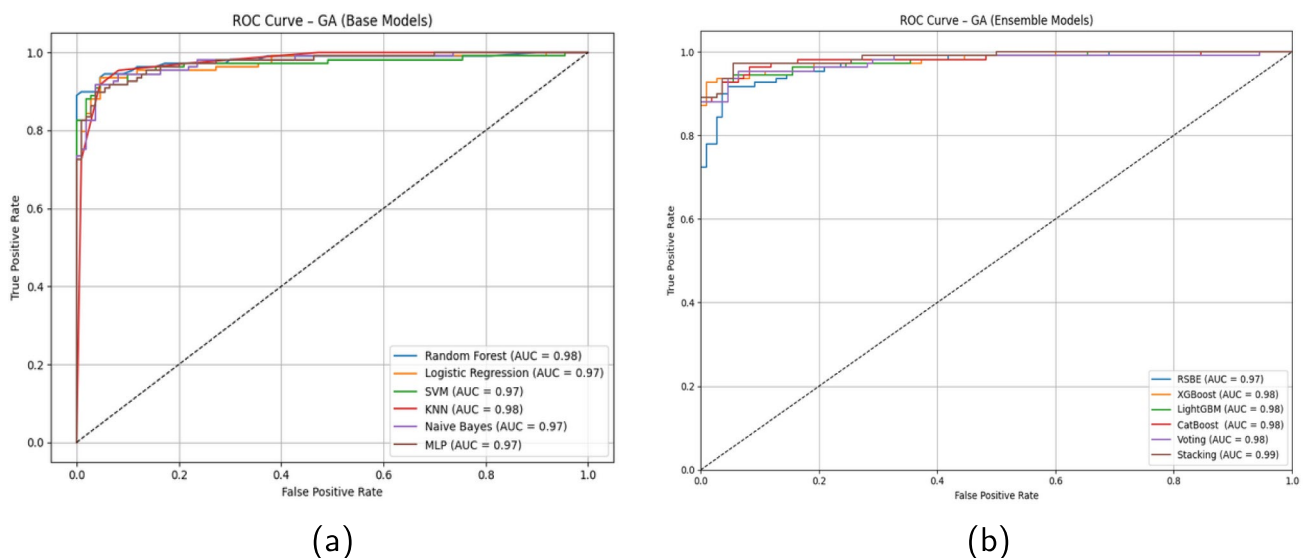


Fig. 6 ROC curves obtained on models built using GA selected features: **a** Base models, **b** Ensemble models

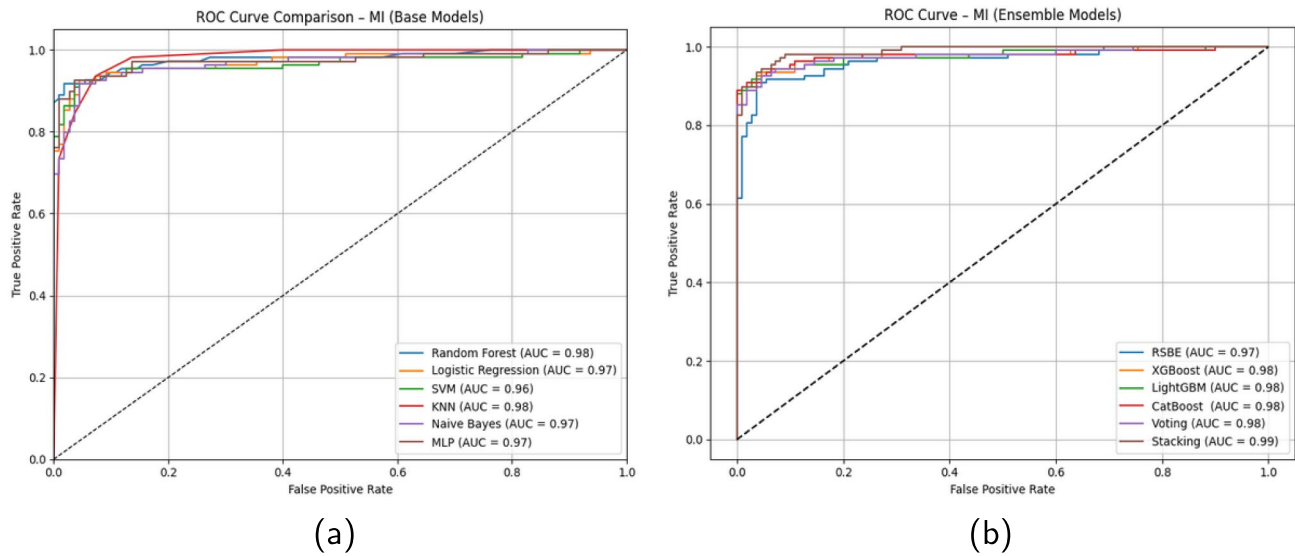


Fig. 7 ROC curves obtained on models built using MI features: **a** Base models, **b** Ensemble models

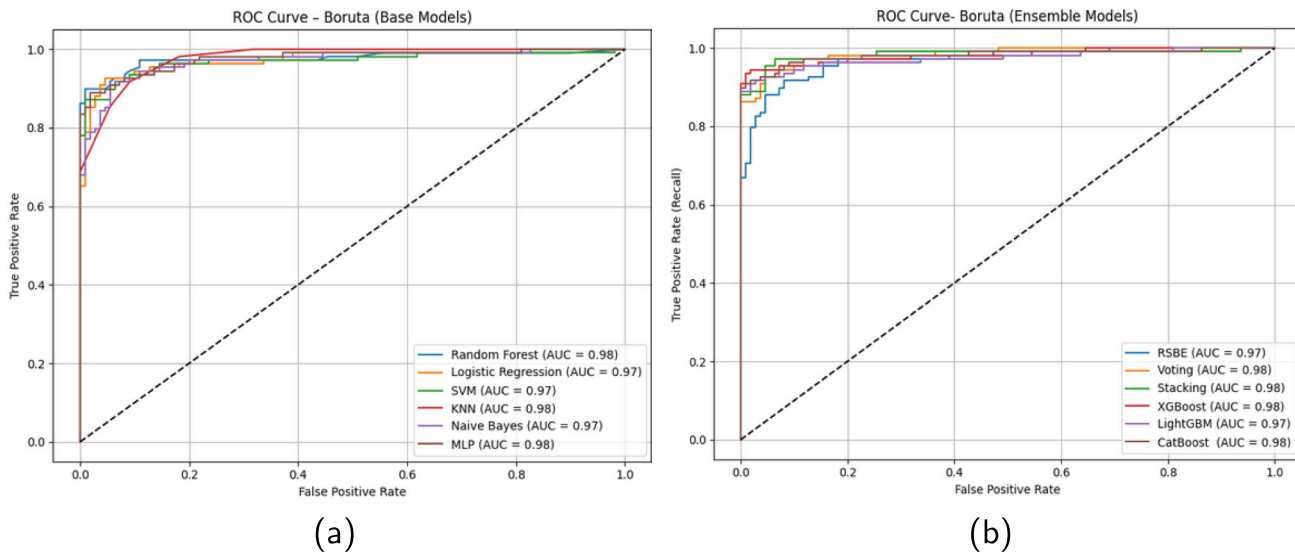


Fig. 8 ROC Curves obtained on models built using Boruta features: **a** Base models, **b** Ensemble models

Table 2 Evaluation of base classifiers—GA selected features

Model	Recall	F-measure	Preci.	Spec.	Acc	AUC
RF	0.8991	0.9289	0.9608	0.9636	0.9315	0.9805
LR	0.9174	0.9346	0.9524	0.9545	0.9361	0.9719
SVM	0.8807	0.9275	0.9796	0.9818	0.9315	0.9701
KNN	0.9541	0.9369	0.9204	0.9182	0.9361	0.9769
NB	0.9174	0.9346	0.9524	0.9545	0.9361	0.9745
MLP	0.8991	0.9202	0.9423	0.9455	0.9224	0.9748

and were evaluated using accuracy, F1-score, precision, specificity, and AUC to ensure a balanced assessment of classification performance. This also made it easier to compare structured with previous PCOS diagnosis research, especially those based on ensemble models.

4.2 Results

Figure 6a, 7 and 8b demonstrate the ROC curves for the base and ensemble models that were produced using each feature selection technique. The robustness of the

Table 3 Improved ensemble model evaluation—GA selected features

Model	Recall	F-measure	Preci.	Spec.	Acc	AUC
RSBE	0.9174	0.9302	0.9434	0.9455	0.9315	0.9736
XGBoost	0.9266	0.9484	0.9712	0.9727	0.9498	0.9813
LightGBM	0.9083	0.9340	0.9612	0.9636	0.9361	0.9788
CatBoost	0.9266	0.9439	0.9619	0.9636	0.9452	0.9811
Voting	0.8807	0.9143	0.9505	0.9545	0.9178	0.9776
Stacking	0.9266	0.9439	0.9619	0.9636	0.9452	0.9870

Table 4 Evaluation of base classifiers—MI selected features

Model	Recall	F-measure	Preci.	Spec.	Acc	AUC
RF	0.8899	0.9327	0.9798	0.9818	0.9361	0.9785
LR	0.9174	0.9346	0.9524	0.9545	0.9361	0.9702
SVM	0.8624	0.9082	0.9592	0.9636	0.9132	0.9647
KNN	0.9358	0.9315	0.9273	0.9273	0.9315	0.9794
NB	0.9083	0.9296	0.9519	0.9545	0.9315	0.9680
MLP	0.8991	0.9333	0.9703	0.9727	0.9361	0.9731

Table 5 Ensemble model evaluation—MI selected features

Model	Recall	F-measure	Preci.	Spec.	Acc	AUC
RSBE	0.9083	0.9296	0.9519	0.9545	0.9315	0.9662
XGBoost	0.9083	0.9384	0.9706	0.9727	0.9406	0.9788
LightGBM	0.8991	0.9333	0.9703	0.9727	0.9361	0.9761
CatBoost	0.9083	0.9340	0.9612	0.9636	0.9361	0.9783
Voting	0.8991	0.9245	0.9515	0.9545	0.9269	0.9768
Stacking	0.9450	0.9493	0.9537	0.9545	0.9498	0.9894

Table 6 Evaluation table—Boruta features (base models)

Model	Recall	F-measure	Preci.	Spec.	Acc	AUC
RF	0.8991	0.9289	0.9608	0.9636	0.9315	0.9760
LR	0.9174	0.9346	0.9524	0.9545	0.9361	0.9717
SVM	0.8716	0.9048	0.9406	0.9455	0.9087	0.9696
KNN	0.9174	0.9132	0.9091	0.9091	0.9132	0.9775
NB	0.9083	0.9209	0.9340	0.9364	0.9224	0.9705
MLP	0.8899	0.9194	0.9510	0.9545	0.9224	0.9766

Table 7 Evaluation table—Boruta features (ensemble models)

Model	Recall	F-measure	Preci.	Spec.	Acc	AUC
RSBE	0.8991	0.9116	0.9245	0.9273	0.9132	0.9690
XGBoost	0.9266	0.9573	0.9902	0.9909	0.9589	0.9819
LightGBM	0.8991	0.9378	0.9800	0.9818	0.9406	0.9740
CatBoost	0.9174	0.9390	0.9615	0.9636	0.9406	0.9816
Voting	0.8991	0.9245	0.9515	0.9545	0.9269	0.9768
Stacking	0.8991	0.9245	0.9515	0.9545	0.9269	0.9827

union-based feature set was demonstrated by the fact that it typically obtained better AUC values across the majority of classifiers.

The GA-selected features provided robust performance in both base and ensemble classifiers. Out of base models, LR, KNN, and NB recorded high and balanced measures with accuracy reaching about 93.61% and an AUC value of 0.9719, 0.9769, and 0.9745, respectively, as reported in Table 2. Results improved further for ensemble models,

where XGBoost, CatBoost, and stacking achieved the highest accuracy values of 94.98%, 94.52%, and 94.52%, respectively. At the same time, these classifiers recorded AUC values of 0.9813, 0.9811, and 0.9870, respectively, as shown in Table 3. In general, GA was effective in improving model generalization without high false positive rates.

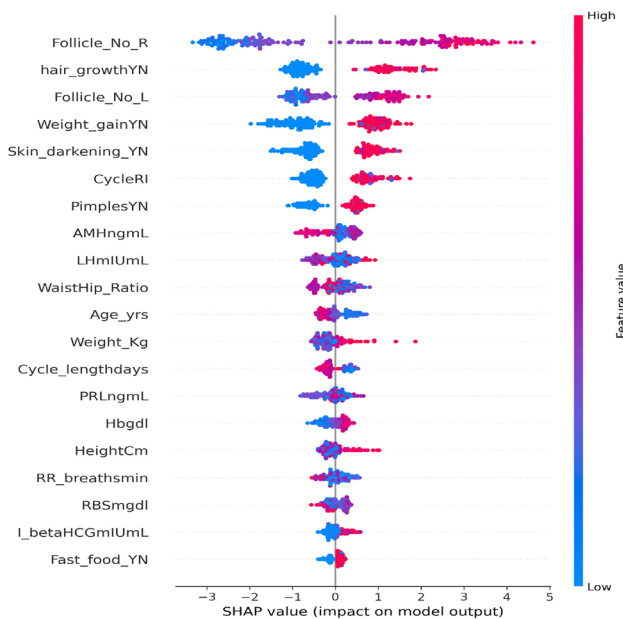
The feature selection using MI enabled many base models to exhibit good performance, as reported in Table 4, with RF achieving the best accuracy (93.61%) and AUC

Table 8 Evaluation table—union features (base models)

Model	Recall	F-measure	Preci.	Spec.	Acc	AUC
RF	0.8991	0.9333	0.9703	0.9727	0.9361	0.9810
LR	0.9083	0.9296	0.9519	0.9545	0.9315	0.9718
SVM	0.8807	0.9231	0.9697	0.9727	0.9269	0.9682
KNN	0.9633	0.9292	0.8974	0.8909	0.9269	0.9756
NB	0.9174	0.9302	0.9434	0.9455	0.9315	0.9730
MLP	0.8807	0.9231	0.9697	0.9727	0.9269	0.9780

Table 9 Evaluation table—union features (ensemble models)

Model	Recall	F-Measure	Preci.	Spec.	Acc	AUC
RSBE	0.8991	0.9202	0.9423	0.9455	0.9224	0.9706
XGBoost	0.9174	0.9434	0.9709	0.9727	0.9452	0.9835
LightGBM	0.8991	0.9202	0.9423	0.9455	0.9224	0.9716
CatBoost	0.9083	0.9340	0.9612	0.9636	0.9361	0.9801
Voting	0.8807	0.9143	0.9505	0.9545	0.9178	0.9776
Stacking	0.9083	0.9252	0.9429	0.9455	0.9269	0.9862

**Fig. 9** Feature contribution interpretation using SHAP

(0.9785). LR and MLP also demonstrated competitive performance, striking a good balance between precision and recall. Ensemble models continued to improve, particularly stacking, and XGBoost achieved high accuracies of 94.98% and 94.06%, respectively. At the same time, both models achieved an AUC of 0.9894 and 0.9788, respectively, as shown in Table 5.

Boruta-selected features allowed base models to possess high overall performance with LR having the best accuracy (93.61%) and RF having the highest precision (96.08%) as given in Table 6. From ensemble models, XGBoost, LightGBM, and CatBoost distinctly outperformed with accuracies of 95.89%, 94.06%, and 94.06%, respectively, and outstandingly high precision and F-measure values as

shown in Table 7. This indicates that Boruta was able to successfully identify the most important predictors for both base and ensemble learners.

With the combination of chosen features, all the base classifiers performed well and consistently, and RF achieved the highest accuracy of 93.61%. At the same time, LR and NB performed with an accuracy above 93.15%. Interestingly, the best recall, 96.33%, was achieved by KNN among the base models, indicating higher sensitivity to positive instances, as shown in Table 8. For the ensemble models, XGBoost once more performed better than others with 94.52% accuracy and an F-measure of 0.9434, and Stacking showed an AUC of 0.9862 as reported in Table 9.

4.3 SHAP-based interpretation of influencing factors

The proposed study utilized SHapley Additive exPlanations (SHAP) values to assess the overall relevance of features in the machine learning model that attained optimal performance. SHAP is a novel approach to making tree-based models more comprehensible. It employs a game-theoretic technique to integrate the local contributions of various features, providing a comprehensive understanding of how the model operates as a whole [39]. The interpretation method employs Shapley values from game theory [40], which allocate contributions equitably among participants. In ML, these values measure the contribution of each feature to a prediction [41, 42], serving as the foundation of the SHAP framework for model interpretability. The proposed work achieved optimum performance in XGBoost with all different feature sets. Therefore, using the prediction results of XGBoost and the union features, the most important contributors to the prediction are assessed using SHAP.

Table 10 Comparison of proposed PCOS detection models with existing works

ML algorithm	Feature selection method	No. of selected features	Splitting ratio	Accuracy (%)	References
RF	Pearson correlation (threshold = 0.8)	38	70:30	93.25	Tiwari et al. [4]
RFLR	Univariate feature selection	10	40-fold cross validation	90.01	Bharati et al. [14]
CatBoost	–	10	20-fold cross validation	93.38	Modi and Kumar [43]
RF	PSO	12	70:30	92.64	R. et al. [31]
RFt	FF	9	70:30	92.02	R. et al. [31]
XGBoost	Chi square & ANOVA	23	80:20	95.83	Khan Inan et al. [33]
Multi-stack of ML	Salp Swarm	15	80:20	98	Khanna et al. [44]
Multi-stack of ML	MI	15	70:30	95	Khanna et al. [44]
XGBoost	GA	23	70:30	94.98	Proposed model
CatBoost, Stacking	GA	23	70:30	94.52	Proposed model
XGBoost	MI	18	70:30	94.06	Proposed model
Stacking	MI	18	70:30	94.98	Proposed model
XGBoost	Boruta	19	70:30	95.89	Proposed model
Light-GBN, CatBoost	Boruta	19	70:30	94.05	Proposed model
XGBoost	Union	29	70:30	94.52	Proposed model

The SHAP beeswarm plot displayed in Fig. 9 offers a comprehensive interpretation of the XGBoost model with the combined feature set, by illustrating how each feature impacts the predictions across all test samples. Features are ordered from most to least influential (top to bottom) according to the mean absolute SHAP value. The primary features, including ‘Follicle_No_R’, ‘hair_growthYN’, ‘Follicle_No_L’, and ‘Weight_gainYN’, play the most significant role in the model’s prediction of the target. The x-axis represents the SHAP value, indicating both the direction and extent of its influence on the prediction. Where, positive SHAP values on the right side influence the positive prediction towards PCOS, and negative SHAP values on the left side exert a downward influence on the prediction, indicating less likelihood of PCOS. Each point in the figure represents a sample, with blue and red indicating low and high values, respectively. This demonstrates that higher

weight, more ovarian follicles, increased AMH levels, and greater hair growth are strong positive indicators of PCOS in a patient. Conversely, lower values for these features generally reduce the likelihood of the PCOS condition. The plot highlights both the influence of each feature on prediction and whether it increases or decreases the risk, making the interpretation straightforward.

4.4 Comparison with existing works

The proposed approach demonstrated a robust and stable performance in various feature selection and ML models compared to the existing work. The performance comparison among these methods is presented in Table 10. In particular, the XGBoost model with Boruta-selected features achieved the highest classification accuracy of 95.89%, recall of 92.66%, precision of 99.02%, F1-score of 95.73%, and AUC of 0.9819. Other strong configurations included MI-based Stacking, with 94.98% accuracy, and a GA-based XGBoost, CatBoost, and Stacking approach, achieving accuracies of 94.98%, 94.45%, and 94.45%, respectively.

In comparison to existing works, the application of Boruta in the proposed approach, a strong and comparatively less-used wrapper technique, offers a valuable and interpretable feature selection path. This makes it easier to make good decisions and makes sure that predictions are accurate.

5 Conclusion

In the proposed work for PCOS detection, different feature selection techniques, namely GA, MI, Boruta, and their union, are analyzed on various ML models. A set of metrics, including accuracy, precision, recall, F1-score, specificity, and AUC, was used to evaluate the performance of the proposed model. The obtained results demonstrated that ensemble classifiers, notably XGBoost, consistently surpassed base classifiers across all feature selection methods. Among all the feature selection techniques, union-based feature selection produced a more comprehensive dataset representation, resulting in improved predictive accuracy with ensemble models. XGBoost achieved the highest classification accuracy of 95.89% using Boruta-selected features, while Stacking demonstrated consistent and superior performance across multiple feature subsets, indicating its generalizability. The outcomes of the proposed work demonstrate that intelligent feature selection, combined with ensemble learning, significantly improves model performance. Feature selection enhances computing efficiency and promotes more efficient model generalization by removing irrelevant or redundant features.

To improve model performance, future work could investigate automatic feature engineering methods, interpretability, augmentation techniques, and more advanced neural architectures. Multimodal data sources can now be further integrated into the proposed feature selection and ensemble learning architecture. Imaging-derived features from ovarian ultrasound scans, in conjunction with sensor-based physiological metrics such as hormone concentrations, pulse rate, and body temperature, can provide a more thorough and detailed perspective of a patient's overall health. Subsequently, the predictive model will improve its clinical accuracy and early detection of PCOS by integrating these textual and visual modalities via deep learning-based fusion methods.

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Data availability The performance of the proposed work was analyzed using the publicly available Kaggle dataset. The dataset is downloaded from <https://www.kaggle.com/datasets/prasoonkottarathil/polycystic-ovary-syndrome-pcos/data>.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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